Supermarket Sales Analysis

Project Report

AICTE | IBM SkillsBuild Data Analytics with AI Internship 2026

In collaboration with BharatCares and IBM

Internship period: 17 August 2026 to 30 September 2026

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1. Abstract

This project analyses 500 supermarket sales transactions recorded from 1 January to 1 July 2026 across four branches in Jaipur, Delhi, Mumbai and Bengaluru. Using Python (pandas, NumPy, Matplotlib and Seaborn) the data is checked for errors, Sales is verified as Quantity x Unit Price, and results are summarised with totals, counts and averages. Charts compare branches, categories, products, customer types and payment methods, and the findings are turned into business decisions. Total sales are 244,411.08 and the average sale is 488.82. Beverages, Personal Care and Dairy lead the categories, and Mumbai is the top branch.

2. Introduction

A supermarket records thousands of sales every month. Turning those records into decisions, such as what to stock, which branch needs support and which customers to reward, is the purpose of data analytics. This project follows the six steps of the Supermarket Sales Analysis project from the internship: load the CSV, check for missing or incorrect values, calculate Sales, group and summarise, create charts, and use the results to make business decisions.

3. Objectives

- Load the sales data and check it for missing or incorrect values.
- Confirm that Sales equals Quantity x Unit Price for every transaction.
- Summarise sales with totals, counts and averages by branch, category, product, customer type, gender, payment method and month.
- Create charts that make the comparisons easy to see.
- Use the results to suggest practical business decisions.

4. Dataset Description

The dataset has 500 rows (one per invoice) and 13 columns. It was provided for the Supermarket Sales Analysis project in the internship (Masterclass 4) and is included with the project files as supermarket_sales.csv.

Column	Description
Invoice ID	Unique invoice number (INV0001 to INV0500)
Date	Date of sale, 1 Jan 2026 to 1 Jul 2026
Branch / City	Branch A Jaipur, B Delhi, C Mumbai, D Bengaluru
Customer Type	Member or Normal
Gender	Male or Female
Product / Category	20 products in 8 categories (Dairy, Grocery, Personal Care, Fruits, Vegetables, Snacks, Beverages, Bakery)
Quantity	Units bought, 1 to 10

Column	Description
Unit Price	Price per unit, 27.67 to 236.71
Payment	UPI, Net Banking, Card or Cash
Rating	Customer rating, 3.0 to 5.0
Sales	Invoice value (Quantity x Unit Price)

5. Tools and Technologies

Tool	Use
Python 3	Programming language
pandas, NumPy	Loading, checking and summarising the data
Matplotlib, Seaborn	Charts and heatmap
Jupyter Notebook	Running and documenting the analysis
IBM SkillsBuild	Learning platform and masterclasses for the internship

6. Methodology

6.1 Data checking

- Loaded the CSV with pandas and parsed the Date column.
- Checked every column for missing values: none were found.
- Checked duplicate rows and duplicate invoice IDs: none were found.
- Checked value ranges for quantity, price, rating and date: all were valid.

6.2 Sales calculation

Sales was recalculated as Quantity x Unit Price and compared with the Sales column. All 500 rows matched, so the data is correct.

6.3 Grouping and summaries

The data was grouped by branch and city, category, product, customer type, gender, payment method and month. For each group the notebook reports total sales, number of transactions, average sale and average rating.

6.4 Charts

Bar charts, a line chart, a pie chart, a histogram and a correlation heatmap were created with Matplotlib and Seaborn.

7. Results

7.1 Key numbers

Measure	Value
Total sales	244,411.08
Transactions	500
Average sale	488.82
Units sold	2,768
Average rating	3.99
Missing values / duplicates	0 / 0

7.2 Sales by branch

Branch	City	Total Sales	Transactions	Avg Rating
C	Mumbai	72,469.45	143	4.05
B	Delhi	64,116.26	133	3.98
D	Bengaluru	55,468.29	119	4.09
A	Jaipur	52,357.08	105	3.84

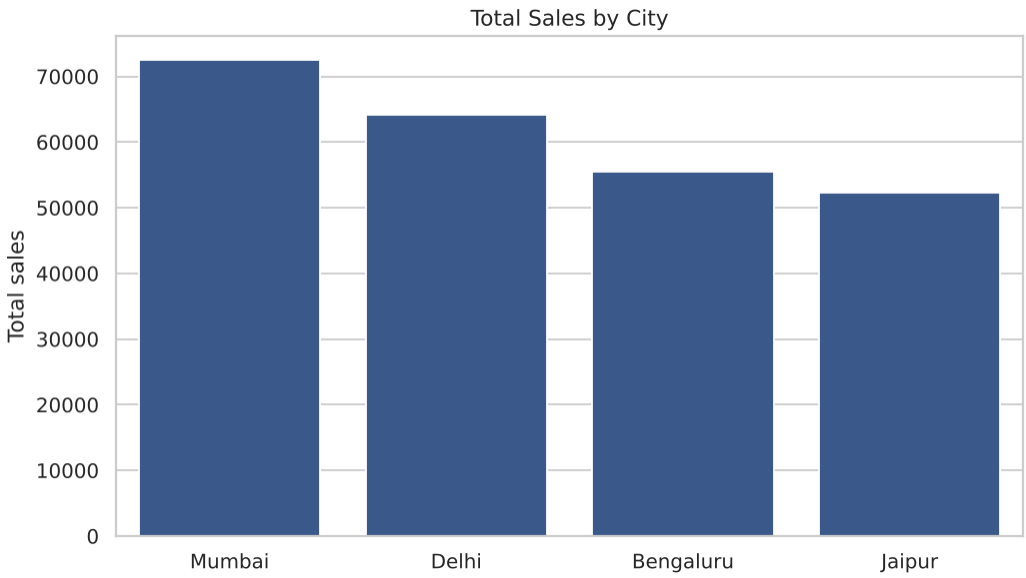


Figure 1: Total sales by city

7.3 Sales by category

Category	Total Sales	Transactions	Avg Sale
Beverages	56,108.24	82	684.25
Personal Care	45,943.96	73	629.37
Dairy	43,992.00	68	646.94
Grocery	40,470.47	82	493.54
Fruits	23,263.17	44	528.71
Snacks	16,992.97	75	226.57

Category	Total Sales	Transactions	Avg Sale
Vegetables	11,124.17	48	231.75
Bakery	6,516.10	28	232.72

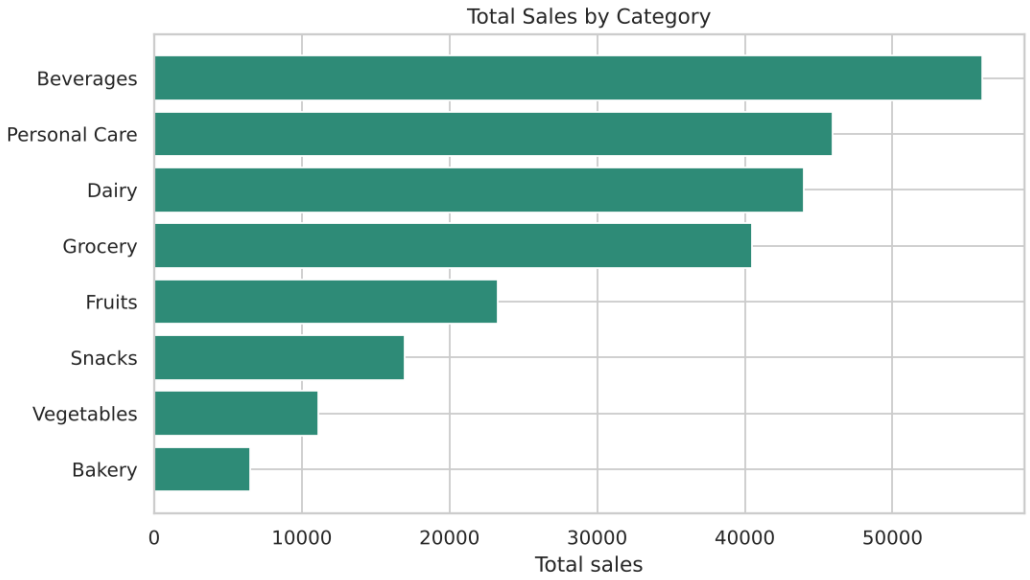


Figure 2: Total sales by category

7.4 Top products

Cheese (27,906), Coffee (27,695) and Shampoo (27,497) are the top three products, followed by Cooking Oil (21,525) and Tea (17,682). Biscuits (3,909) and Tomato (4,325) sell the least.

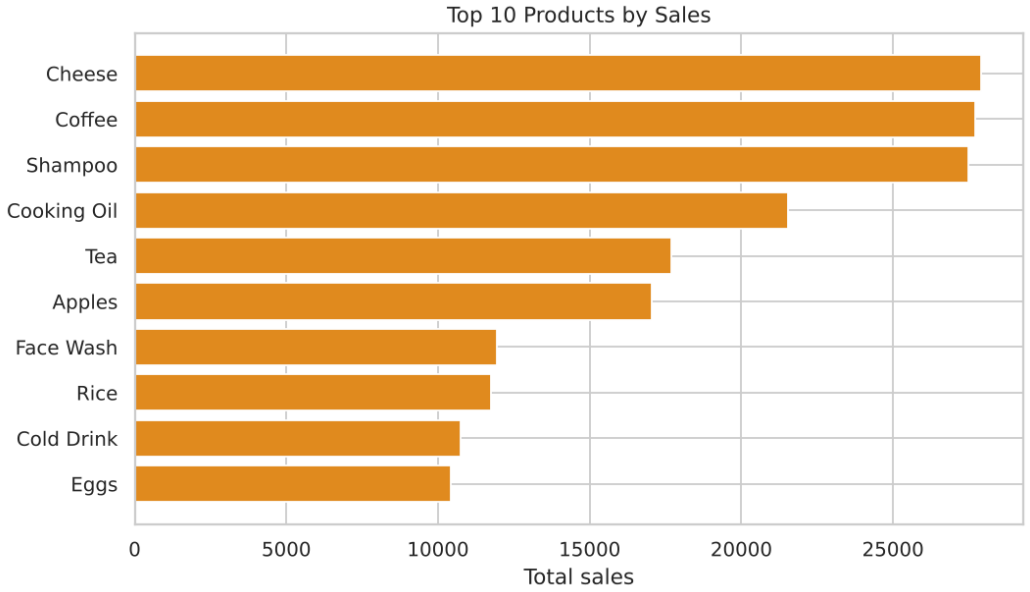


Figure 3: Top 10 products by sales

7.5 Monthly trend

Sales were highest in April (52,570) and lowest in February (30,068) among full months. July has only 7 transactions because the data ends on 1 July 2026, so it is a partial month and should not be compared with the others.

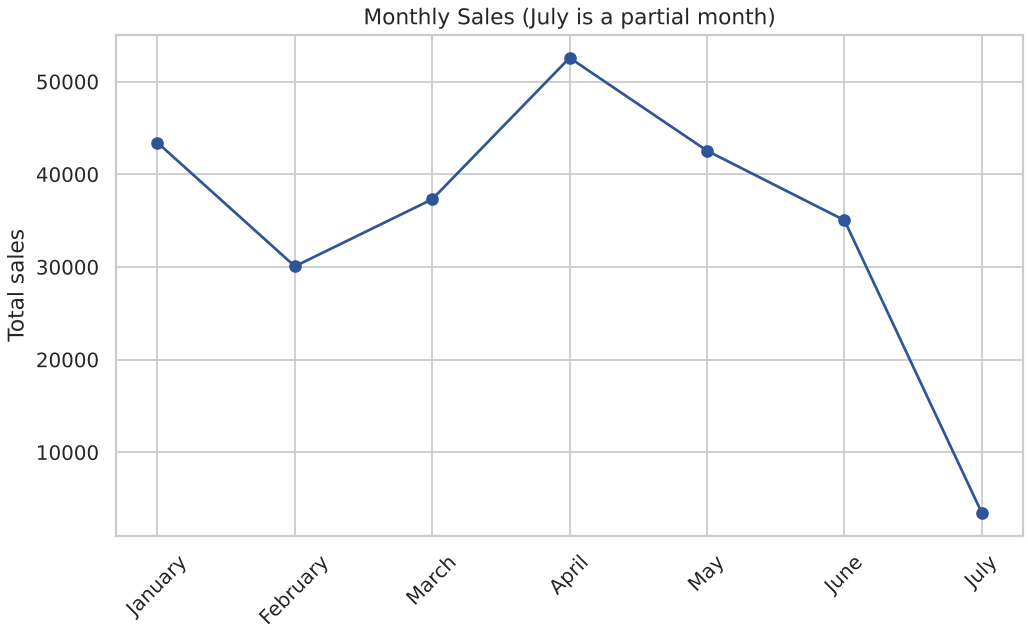


Figure 4: Monthly sales

7.6 Customer type, gender and payment

Members made 296 purchases worth 143,009 (about 59% of sales) and Normal customers made 204 purchases worth 101,402. Female and male customers spent almost the same (123,954 and 120,457). Payments are spread evenly across UPI, Net Banking, Card and Cash (about 25% each), but UPI (67,910) and Net Banking (65,195) bring the highest sales, while Cash has the lowest average sale (443).

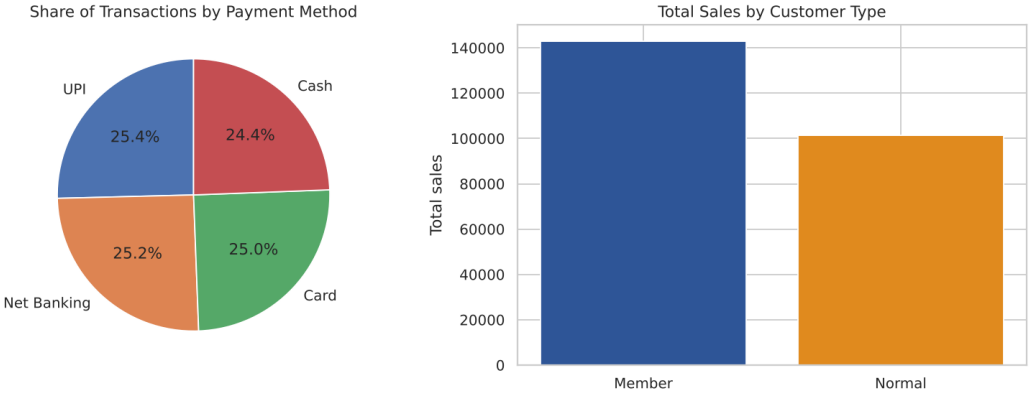


Figure 5: Payment method share and sales by customer type

7.7 Ratings and correlation

The average rating is 3.99 and ratings range from 3.0 to 5.0. Bakery (4.24) has the best average rating and Fruits (3.83) the lowest. Sales has a correlation of 0.72 with Unit Price and 0.57 with Quantity, so price affects sales more than quantity. Rating has almost no relation to sales (-0.05).

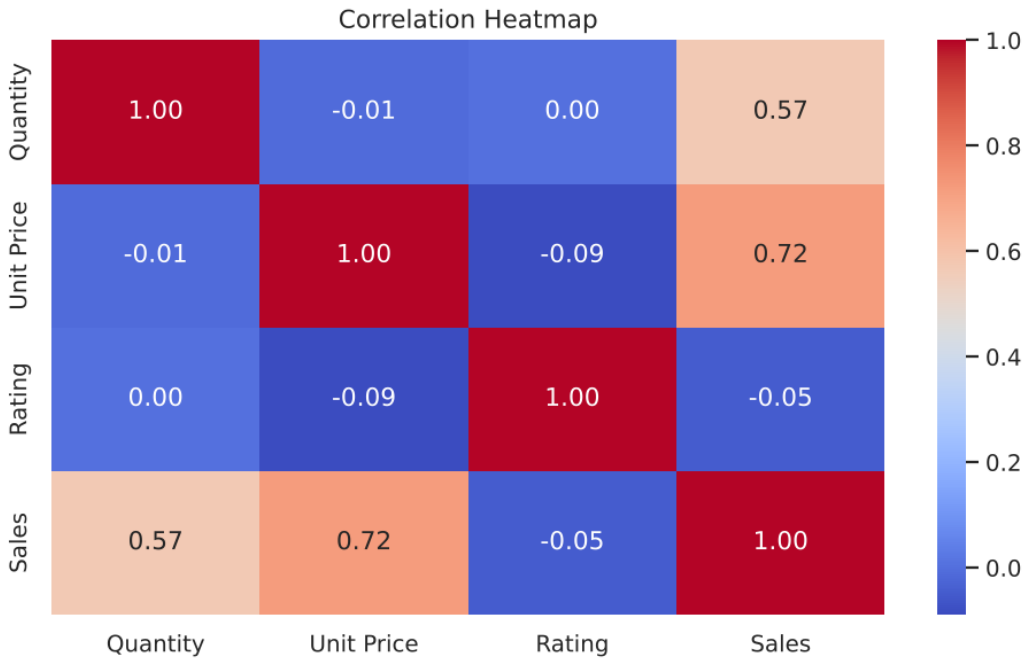


Figure 6: Correlation heatmap

8. Business Decisions

- Focus on Beverages, Personal Care and Dairy, which bring the most sales, and keep Cheese, Coffee and Shampoo well stocked.
- Raise basket value in Snacks and Bakery, where many small sales give low totals, with bundle offers and add-on deals.
- Support Branch A (Jaipur), which has the lowest sales and the lowest rating (3.84); review its service and stock.
- Promote UPI and Net Banking offers, and encourage Cash customers to switch, since Cash has the lowest average sale.
- Reward members with member-only offers, because they bring about 59% of sales.
- Give priority to premium products, because price drives sales more than quantity.

9. Limitations

- Only 500 transactions from 1 January to 1 July 2026 are included, and July is incomplete.
- Only four branches are covered, so results may not apply to other stores.
- The analysis is descriptive. It does not predict future sales or prove why differences occur.

10. Future Scope

- Forecast monthly sales with time-series models once more data is available.
- Group customers with clustering for targeted offers.
- Build an interactive dashboard for store managers.
- Use an AI assistant such as IBM Bob to generate written summaries from the results.

11. Conclusion

The dataset is clean, and the analysis shows that sales are driven by premium products in Beverages, Personal Care and Dairy and by the Mumbai and Delhi branches. Members are the most valuable customer group, and price matters more than quantity. The project applied the internship steps of data checking, summarising, visualising and decision making from start to finish, and it can be extended with forecasting and dashboards.

12. References

- IBM SkillsBuild Data Analytics with AI masterclasses and the Supermarket Sales Analysis DA project workbook, BharatCares / AICTE, 2026.
- pandas, Matplotlib and Seaborn official documentation.